

4 The matrix $\mathbf{M}$ is $\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

(a) (i) Calculate $\det \mathbf{M}$. [1]

(ii) State two geometrical consequences of this value for the transformation associated with $\mathbf{M}$. [2]

(b) Describe fully the transformation associated with $\mathbf{M}$. [1]